

TRIGGERS TO STAR FORMATION

Clouds of interstellar material need a trigger to start them collapsing, since they are held up by their own pressure and that of internal magnetic fields. Such a trigger might be as simple as the gravitational tug from a passing star, or it might be a shock wave caused by the blast from a supernova or the collision of two or more galaxies. In spiral galaxies such as the Milky Way, density waves move through the dust and gas in the galactic disk (see p.227). As the waves pass, they temporarily increase the local density of interstellar material, causing it to collapse. Once the waves have passed, their shape can be picked out by the trails of bright young stars.



GALACTIC COLLISIONS

A ring of stars is created when two galaxies collide. Here, shock waves have rippled out, triggering star formation in the interstellar material.

FROM OLD TO NEW

Shock waves and material from a supernova blast spread out through the interstellar medium, triggering new star formation.



STAR CLUSTERS

When they have formed from the fragmentation of a single collapsing molecular cloud, young stars are often clustered together. Many stars are formed so close to their neighbors that they are gravitationally bound, and some are even close enough to transfer material. It is unusual for a star not to be in a multiple system such as a binary pair (see pp.274–275), and in this respect, the Sun is uncommon. Stars within a cluster usually have a similar chemical composition, although, since successive generations of stars may be produced by a single nebula, clusters may contain stars of different ages (see pp.288–289). Remnants of dust and gas from the initial cloud will linger, and the dust grains often reflect the starlight, predominantly in the shorter blue wavelengths.

VIOLENT STAR FORMATION

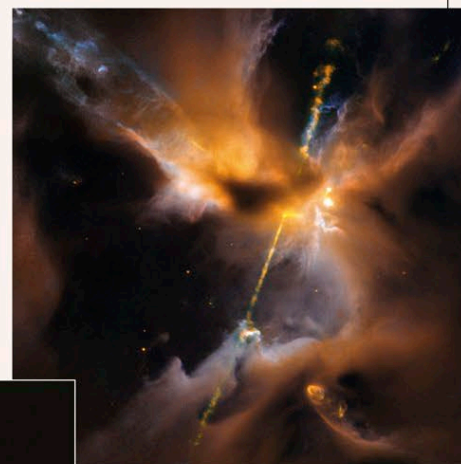
Young star clusters (blue) and star-forming regions (pink) abound in NGC 1427A. As the galaxy's gas collides with the intergalactic medium through which the galaxy is traveling, the resulting pressure triggers violent but stunning star-cluster formation.

Thus, young star clusters are often surrounded by distinctive blue reflection nebulae. Young stars are hot and bright, and any nearby interstellar material will be heated by new stars' heat, producing red emission nebulae. Stars' individual motions will eventually cause a young star cluster to dissipate, though multiple stellar systems may remain gravitationally bound and may move through a galaxy together.

TOWARD THE MAIN SEQUENCE

As collapsing fragments of nebulae continue to shrink, their matter coalesces and contracts to form protostars. These stellar fledglings release a great deal of energy as they continue to collapse under their own gravity. However, they are not easily seen because they are generally surrounded by the remnants of the cloud from which they formed. The heat and pressure generated within protostars acts against the gravity of their mass, opposing the collapse. Eventually, matter at the centers of the protostars gets so hot and dense that

nuclear fusion starts and a star is born. At this stage, stars are very unstable. They lose mass by expelling strong stellar winds, which are often directed in two opposing jets channeled by a disk of dust and gas that forms around their equators. Gradually, the balance between gravity and pressure begins to equalize and the stars settle down on to the main sequence (see pp.234–237).

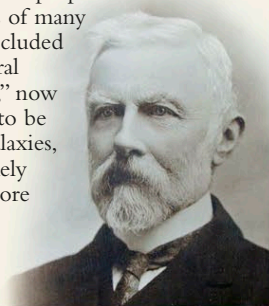


ADOLESCENT STAR

As young stars gather more material onto their equators, they eject excess matter from their pole (left). This can produce twin beams of glowing gas known as Herbig-Haro objects. HH 24 (above) is one spectacular example.

J. L. E. DREYER

Danish-Irish astronomer John Louis Emil Dreyer (1852–1926) compiled the *New General Catalog of Nebulae and Clusters of Stars*, from which nebulae and galaxies get their NGC number. At the time of compilation, it was not known if all the nebulous objects were within the Milky Way. Dreyer studied the proper motions of many and concluded the “spiral nebulae,” now known to be spiral galaxies, were likely to be more distant objects.



polar
gas jets
accretion
disk

